Thm. Let $f:[a,b]\times[c,d]\to\mathbb{R}$ be a continuous map. Define
$$\phi(x)=\int_c^d f(x,t)\,dt,\qquad x\in[a,b].$$
Then, the following properties hold:

1) ϕ is continuous.

2). If the partial derivative $D_x f$ exists and continuous on $(a,b)\times[c,d]$, then ϕ is differentiable on (a,b) with the derivative is
$$\phi'(x)=\int_c^d D_x f(x,t)\,dt,\qquad x\in(a,b).$$

proof. Since $[a,b]\times[c,d]$ is compact, f is continuous uniformly on the domain. Therefore, given $\varepsilon>0$, there exists $\delta>0$ such that:
$$|x-y|<\delta,\ t\in[c,d]\ \text{implies}\ |f(x,t)-f(y,t)|<\frac{\varepsilon}{d-c}.$$
Now, by the property of integral, $|x-y|<\delta$ implies
$$|\phi(x)-\phi(y)|=\left|\int_c^d f(x,t)-f(y,t)\,dt\right|\le\int_c^d|f(x,t)-f(y,t)|\,dt<(d-c)\cdot\frac{\varepsilon}{d-c}=\varepsilon.$$
So, the first assertion is proved. To show the second assertion, observe that
$$\int_x^{x+h} D_x f(z,t)\,dz=f(x+h,t)-f(x,t)$$
by the Fundamental Theorem of Calculus, since the $D_x f$ is continuous. Now,
$$\frac{1}{h}\cdot[\phi(x+h)-\phi(x)]=\frac{1}{h}\cdot\int_c^d f(x+h,t)-f(x,t)\,dt=\frac{1}{h}\int_c^d\int_x^{x+h}D_x f(z,t)\,dz\,dt$$
where $h\neq0$, and $x\in(a,b)$ was arbitrary. Take $r>0$ such that $x\in[x-r,x+r]\subseteq(a,b)$. Then, $D_x f$ is continuous uniformly on $[x-r,x+r]\times[c,d]$, so that given $\varepsilon>0$, there exists $\delta>0$ such that
$$|z-x|<\delta,\ t\in[c,d]\ \text{implies}\ |D_x f(z,t)-D_x f(x,t)|<\frac{\varepsilon}{d-c}.$$
Finally,
$$\left|\frac{\phi(x+h)-\phi(t)}{h}-\int_c^d D_x f(x,t)\,dt\right|=\left|\int_c^d\frac{1}{h}\cdot\int_x^{x+h}D_x f(z,t)-D_x f(x,t)\,dz\,dt\right|$$
$$\le\int_c^d\frac{1}{h}\cdot\int_x^{x+h}|D_x f(z,t)-D_x f(x,t)|\,dz\,dt<\varepsilon.\ \blacksquare$$